

PROJECT TOPIC

ANALOG AND MIXED-SIGNAL IC DESIGN

❖ List of devices

Type	Library	Name	L_unit	W_unit	Note
NMOS	gpdk045	nmos1v nmos1v_lvt nmos1v_hvt	45n	120n/150n	
PMOS	gpdk045	pmos1v pmos1v_lvt pmos1v_hvt	45n	120n/150n	
RES	gpdk045	ressnpoly	x	Segment Width = 1um	
MOSCAP	gpdk045	nmoscap1v pmoscap1v	x	x	
MIMCAP	gpdk045	mimcap	x	x	
BJT	gpdk045	nnp2/vpnp2 nnp5/vpnp2 nnp10/vpnp2	x	x	
DIODE	gpdk045	ndio/pdio ndio/pdio_lvt ndio/pdio_hvt	x	x	
INDUCTOR	gpdk045	ind_a ind_s	x	x	

- Students are not allowed to use any ideal devices (from ahdlLib, analogLib, ...) for the project

❖ List of std cells: Using devices in gsclib045/gsclib045_lvt/gsclib045_hvt library

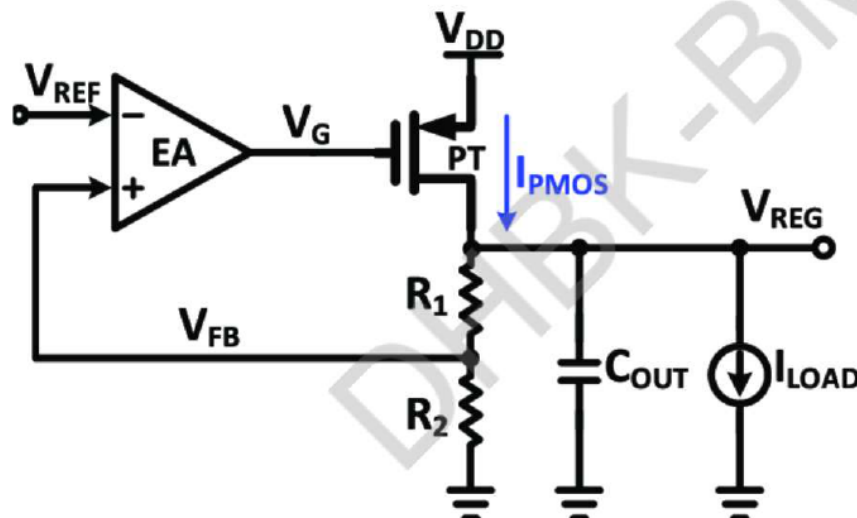
1. Low Dropout Voltage (LDO)

✓ **Specifications:** Design basic LDO (students will choose architecture or use proposed architecture)

- PVT:
 - Temperature: -40°C , 25°C , 125°C
 - Process: FF, SS, TT
- Input (supply): 1.1V
- Output: $0.75\text{V} \pm 5\%$
- Reference voltage: 0.6V
- Output current load: 10u to 1mA

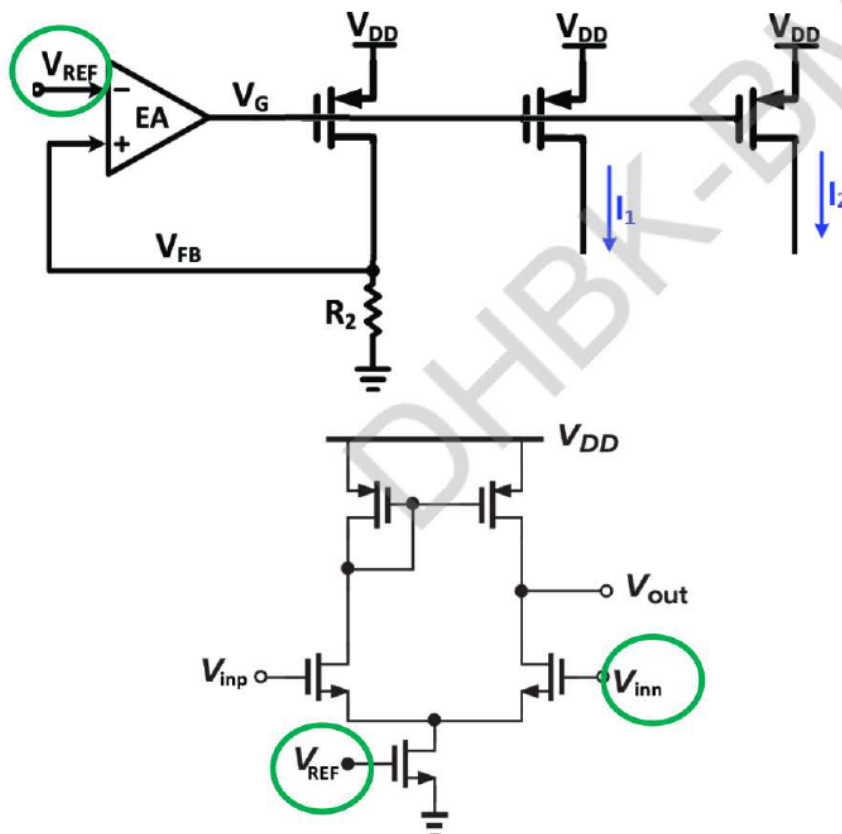
✓ **Requirements:** Students will build architecture, schematic, simulation testbench on cadence tools, study (understand) and can measure below parameters.

- DC: **Output voltage, current load**, MOS operation region (V_{od} , $V_{\text{sat,margin}}$)
- AC: **phase margin** > 60deg, **DC gain** > 20dB, 3dB-BW. (Using LSTB analysis)
- Transient: ideal switching load, real load assumption with an inverter
- PSRR@DC < -30dB, PSRR max < 0dB



2. Current Generator

- ✓ **Specification:** Design a current generator (students will choose architecture or use proposed architecture)
 - PVT:
 - Temperature: -40°C , 25°C , 125°C
 - Process: FF, SS, TT
 - Input (supply): 1.1V
 - Reference voltage: 0.75V
 - Output current: $I_1 = 100\mu \pm 3\%$, $I_2 = 200\mu \pm 5\%$
- ✓ **Requirements:** Students will build architecture, schematic, simulation test bench on cadence tools, study (understand) and can measure below parameters.
 - DC: **Output current**, MOS operation region (V_{od} , $V_{sat,margin}$)
 - AC: **phase margin** $> 60\text{deg}$, **DC gain** $> 20\text{dB}$, 3dB-BW. (Using LSTB analysis)
 - Transient: resistive load, current mirror load
 - **PSRR@DC** $< -20\text{dB}$



3. Flash ADC

- ✓ **Specification:** Design flash ADC (students will choose architecture or use proposed architecture)

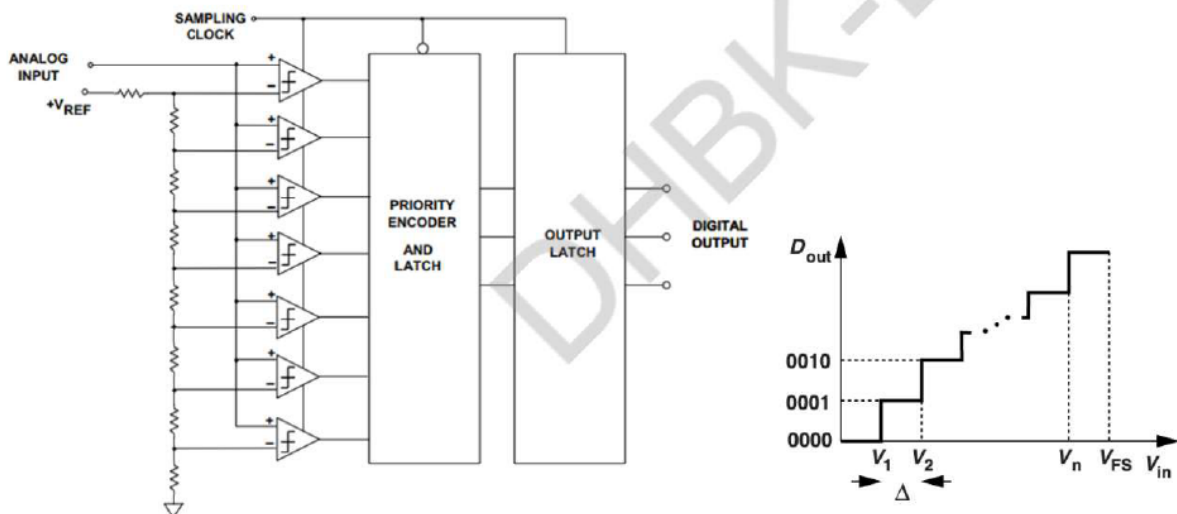
- PVT:
 - Temperature: -40°C , 25°C , 125°C
 - Process: FF, SS, TT
- Supply voltage: 1.1V
- Reference voltage: 0.55V
- Output digital: 3 bits.

- ✓ **Requirements:** Students will build architecture, schematic, simulation test bench on cadence tools, study (understand) and can measure below parameters.

- There are two types to build the comparators:
 - **Inverter:** verify trip point if using an inverter for the comparator
 - **Opamp:** verify dc operating point if using opamp
- **Truth table verification** (with schematic) to confirm the correct function.

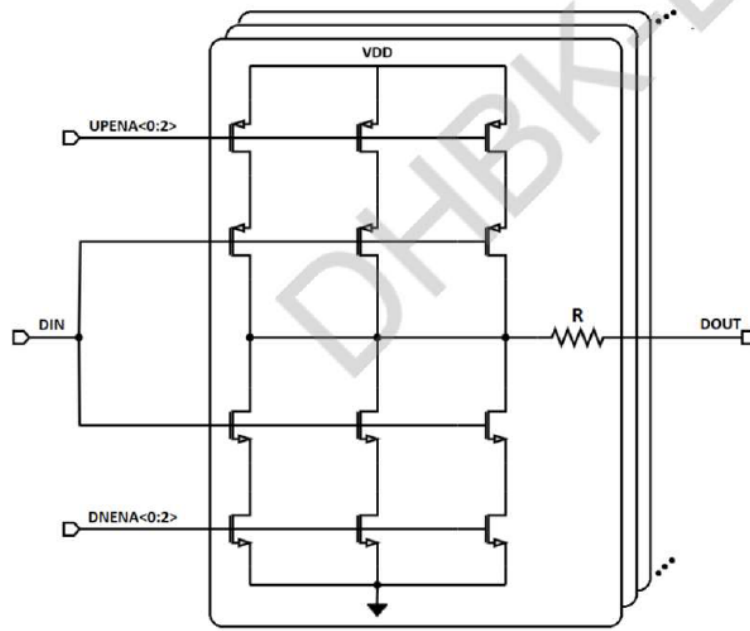
Error of switching voltage $< 5\text{mV}$

- Measure $\text{DNL} < 0.2 \cdot \text{LSB}$, $\text{INL} < 0.2 \cdot \text{LSB}$, $\text{SNDR} > 24\text{dB}$, $\text{ENOB} > 3.5\text{bits}$, Power
- Transient simulation with sine wave, ramp wave



4. Transmitter Driver (TX)

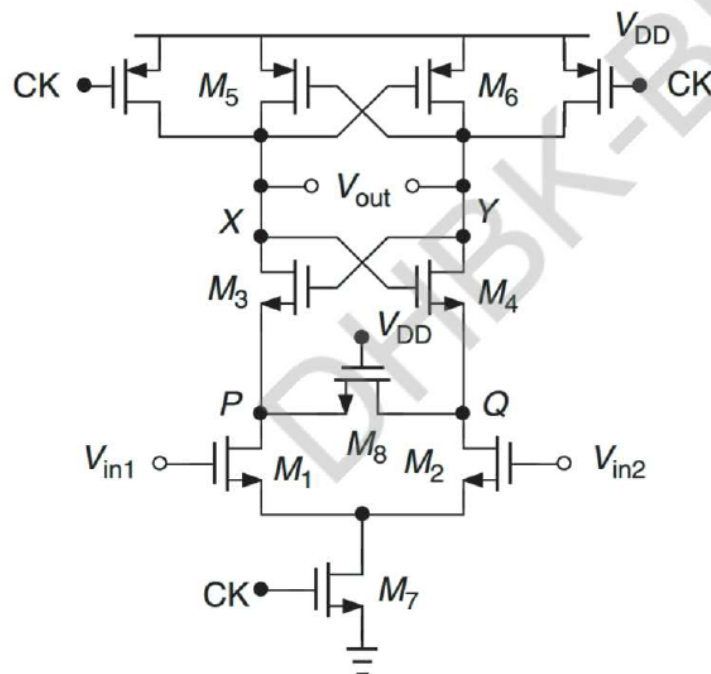
- ✓ **Specification:** Design a driver (students will choose architecture or use proposed architecture)
 - Supply voltage: 0.9V
 - Impedance control: 3 bits
 - **Output impedance: 50 ohms**
 - PVT:
 - Process: FF, SS, TT, **SF, FS**
 - Temp: -40°C, 25°C, 125°C
 - Voltage: Supply voltage +/-5%
 - **Operation speed: 5Gbps, Output cap (connect to Dout): 100fF**
- ✓ **Requirements:** Students will build architecture, schematic, simulation test bench on cadence tools, study (understand) and can measure below parameters.
 - Measure **output impedance** (pull up/pull down), **impedance range and step**
 - **Measure total output capacitance**
 - Transient simulation with clock, prbs data (can be built by vbit) source:
measure **rise/fall time < 25ps** , **duty cycle distortion < ±3%**
 - Measure jitter after calibration



12 slices

5. Strong arm latch (SA)

- ✓ **Specification:** Design a strong-arm latch (students will choose architecture or use proposed architecture)
 - Supply voltage: 1.0V
 - PVT:
 - Temperature: -40°C , 25°C , 125°C
 - Process: FF, SS, TT
 - Input voltage: peak to peak = 100mV
 - Input voltage common: supply voltage/2
 - Operation speed: 5Gbps, Output cap (connect to X/Y): 10fF
- ✓ **Requirements:** Students will build architecture, schematic, simulation test bench on cadence tools, study (understand) and can measure below parameters.
 - **Correct function @5Gbps.** Explain the result
 - Transient simulation with clock, prbs source: measure rise/fall time, duty cycle, **latency from input clock to output response**, Power
 - **Analyze and verify to improve SA latch latency < 20ps**



- Reference paper: https://www.seas.ucla.edu/brweb/papers/Journals/BR_Magzine4.pdf

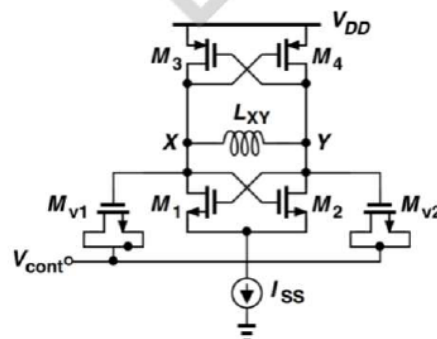
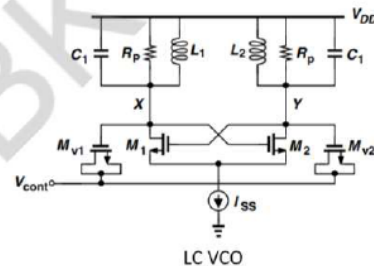
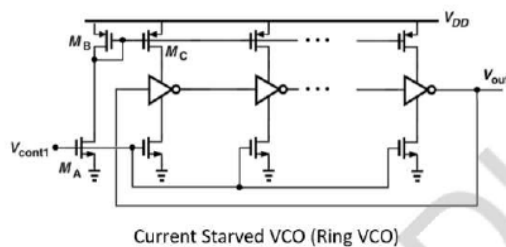
6. Voltage Control Oscillator (VCO)

- ✓ **Specification:** Design a VCO (students will choose architecture or use proposed architecture).

- PVT:
 - Temperature: -40°C , 25°C , 125°C
 - Process: FF, SS, TT
 - Supply: $1.1\text{V} \pm 5\%$
- **Input voltage control level: $0.3 \cdot V_{DD}$ to $0.7 \cdot V_{DD}$**
- **Output target frequency: $5\text{GHz} @ V_{DD}/2$**

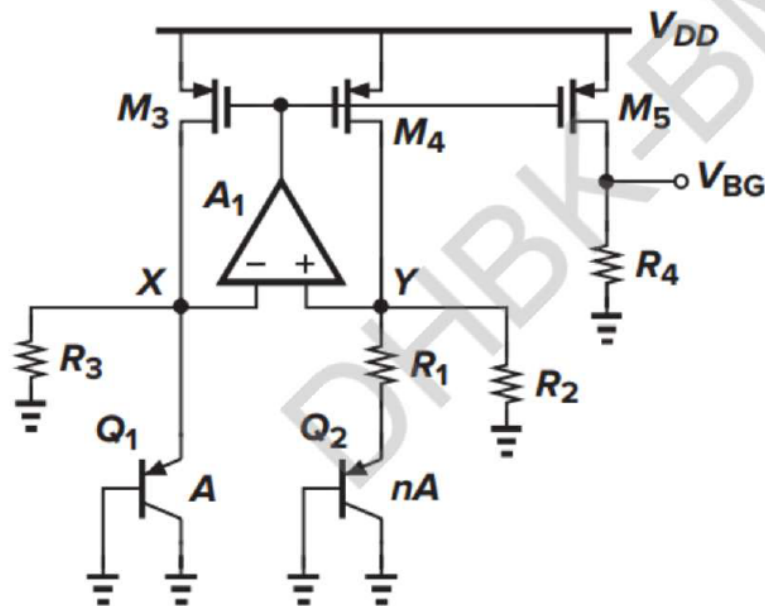
- ✓ **Requirements:** Students will build architecture, schematic, simulation testbench on cadence tools, study (understand) and measure below parameters.

- The output response of VCO with target frequency using transient analysis $5\text{GHz} @ V_{DD}/2$. Measure the output **rising/falling time $< 0.1 \cdot UI$ and duty cycle distortion $< \pm 3\%$**
- The tuning range of VCO with control voltage from $0.3 \cdot V_{DD}$ to $0.7 \cdot V_{DD}$ (**tuning range can be covered 5GHz the variation of PVT**)
- **Jitter $< 5\text{ps}$** – Using eye diagram.
- **Phase noise $< -80\text{dBc/Hz}$** – Using pss and pnoise analysis.
- Power.



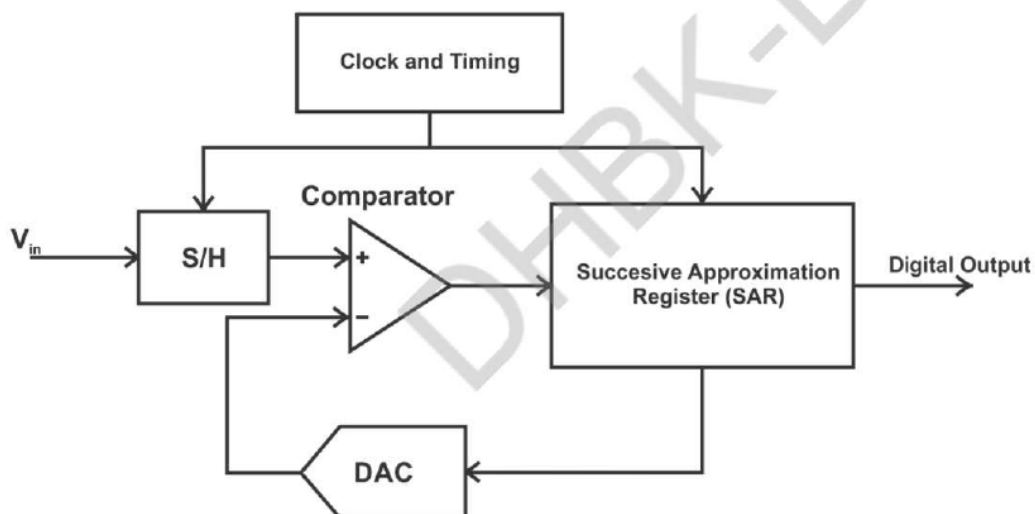
7. Bandgap Reference (BGR)

- ✓ **Specification:** Design a BGR (students will choose architecture or use proposed architecture).
 - PVT:
 - Supply voltage: 1.1V $\pm$ 5%;
 - Temperature: $-40^{\circ}\text{C} \rightarrow 125^{\circ}\text{C}$
 - Process: FF, SS, TT
 - **Output voltage:** 0.75V $\pm$ 1%
- ✓ **Requirements:** Students will build architecture, schematic, simulation testbench on cadence tools, study (understand) and measure parameters below.
 - DC: **Output voltage**, MOS operation region (V_{od} , $V_{sat,margin}$)
 - **Temperature Coefficient** $< 60\text{ppm/C}$ (DC curves vary with Temp)
 - **Gain** $> 20\text{dB}$ and **Phase margin** $> 60\text{deg}$ – using LSTB analysis
 - **PSRR_DC** $< -20\text{dB}$
 - **Line Regulator** $< 10\text{mV/V}$



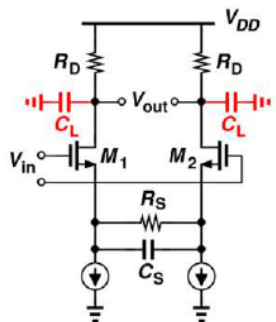
8. Successive Approximation Register ADC (SAR ADC)

- ✓ **Specification:** Design SAR ADC (students will choose architecture or use proposed architecture)
 - PVT:
 - Supply voltage: 1.1V
 - Temperature: -40°C , 25°C , 125°C
 - Process: TT
 - **Output digital (resolution): 8 bits**
 - **Sampling rate: 1MSps** (can improve to work with 10MSps / 50MSps / 100MSps...)
- ✓ **Requirements:** Students will build architecture, schematic, simulation test bench on cadence tools, study (understand) and can measure below parameters.
 - **Truth table verification** (with schematic) to confirm the correct function.
 - **Error of switching voltage $< 2\text{mV}$**
 - Verify and check the performance and function of each sub-block
 - Measure **DNL $< 0.2 \cdot \text{LSB}$, INL $< 0.2 \cdot \text{LSB}$, SNDR $> 45\text{dB}$, ENOB $> 7\text{bits}$, Power**
 - Transient simulation with sine wave, ramp wave

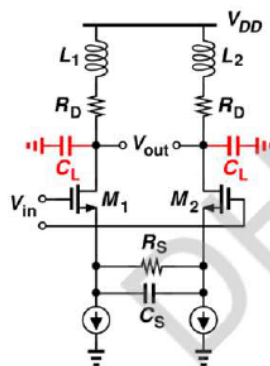


9. Continuous time linear equalization (CTLE)

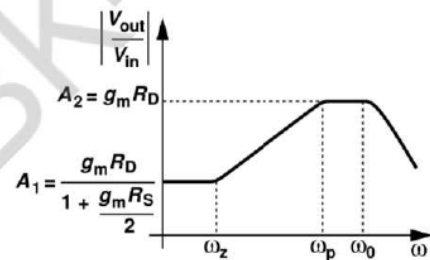
- ✓ **Specification:** Design a CTLE (students will choose architecture or use proposed architecture)
 - PVT:
 - Supply voltage: 1.1V+/-5%;
 - Temperature: -40°C, 25°C, 125°C
 - Process: TT, FF, SS
 - **Input common mode voltage: supply voltage/2**
 - Reference current (IREF): 50uA, build current mirror topology.
 - **Load capacitor: 20Ff**
 - **Operation speed: 5Gbps**
- ✓ **Requirements:** Students will build architecture, schematic, simulation test bench on cadence tools, study (understand) and can measure below parameters.
 - AC analysis: **Measure DC Gain, Peaking Gain, zeros and poles in AC response.** Calculate in theory, implement and verify function @5Gbps.
 - Transient analysis: **transient response of CTLE (slew time, duty cycle)**
(Create a channel loss -20db@5Gbps to the check performance of CTLE)
 - **Plot eye diagram and measure jitter.**



(a) Basic CTLE architecture



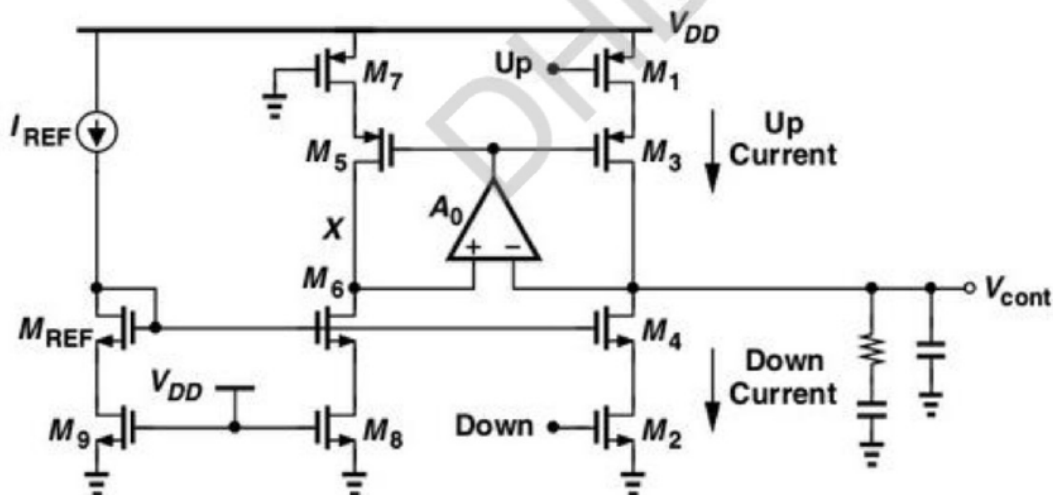
(b) CTLE with peaking inductor architecture



(c) CTLE frequency response

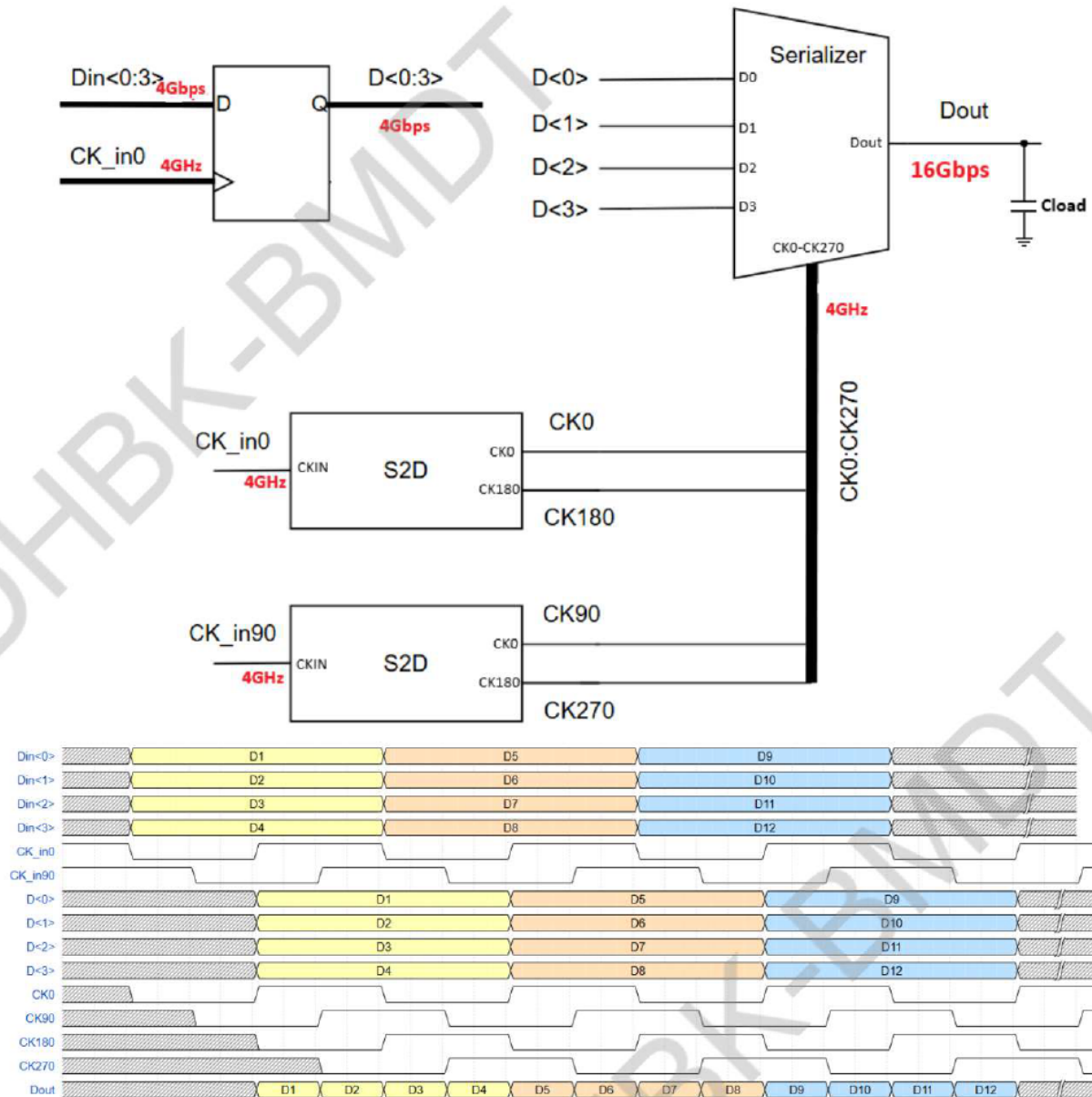
10. Charge pump for PLL (CP)

- ✓ **Specification:** Design a charge pump for PLL (students will choose architecture or use proposed architecture)
 - PVT:
 - Supply voltage: 1.1V
 - Temperature: -40°C , 25°C
 - Process: TT, FF, SS, **SF, FS**
 - Reference current (I_{REF}): 50uA.
 - Current output: $I_{\text{UP}} = I_{\text{DN}} = 50\mu\text{A} \pm 5\%$;
 - **Loading:** $R_1 = 150\text{k}\Omega$, $C_1 = 3.5\text{pF}$, $C_2 = 0.35\text{pF} - (R_1 + C_1) \parallel C_2$
 - Operation speed UP/DN: 10MHz. Minimum pulse width high of UP/DN: 20ns
- ✓ **Requirements:** Students will build architecture, schematic, simulation testbench on cadence tools, study (understand) and measure parameters below.
 - DC: MOS operation region (V_{od} , $V_{\text{sat,margin}}$), **Current mismatch between UP and DN < 3% when V_{cont} varies from $0.3 \cdot V_{\text{DD}}$ to $0.7 \cdot V_{\text{DD}}$**
 - **Gain > 20dB and Phase margin > 60deg – using LSTB analysis**
 - Transient analysis: Set up the testbench with UP/DN at 10 MHz, a delay phase of 50% UI, and a pulse width (high) of 20 ns for UP/DN.
 - Check the range of V_{cont} and the ripple on V_{cont} .
 - Increase pulse width of UP to 50ns. Check the range of V_{cont} and the ripple on V_{cont} . Redo the same for pulse width of DN.



11. Serializer and S2D (SER)

- ✓ **Specification:** Design an 16Gbps Serializer (students will use proposed architecture)
 - PVT:
 - Supply voltage: 1.1V;
 - Temperature: -40°C , 25°C
 - Process: TT, FF, SS, **SF, FS**
 - Input data: **Din<0:3>** as independent data streams at **4Gbps**, **rise/fall times = 1.5ps**. Set the **phase skew between data paths to 5ps**, Din<3> is MSB.
 - Input clock: **CK_in0** and **CK_in90** are a **4 GHz** with a 50% duty cycle **with 0° and 90° phase, respectively, means phase shift 125ps**; rise/fall times = 2ps
 - Output loading: $C_{load} = 20fF$
 - Assume the **CK_in0** and **CK_in90** are aligned and there is no phase deskew.
- ✓ **Requirements:** Students will build architecture, schematic, simulation testbench on cadence tools, study (understand) and measure parameters below.
 - Correct the function of diagram: Output FF D<0:3>, Output S2D CK0:CK270 and output Dout.
 - Verify setup time and hold time of FF and Serializer.
 - **Phase difference between CK0/CK90/CK180/CK270 < 1.5ps**
 - Output transient performance:
 - **Rising time/ Falling time < 10ps**
 - **Duty cycle distortion < $\pm 5\%$**
 - Output swing: min < $0.1 \cdot VDD$, max > $0.9 \cdot VDD$
 - Measure Power of each block.
- ✚ This architecture can be violated the timing analysis. Students need to check carefully all data and clock paths.



- Propose S2D design:

